Teyzix Core Internship (June Batch)

Task Title: Customer Behavior Analytics & Churn Prediction Dashboard
Task ID: ML-INT-1
Domain: Machine Learning
Difficulty Level: Intermediate
Assigned Date: 11 June 2026
Submission Deadline: 19 June 2026

Objective
Build an end-to-end system that analyzes customer behavior data, generates business insights, and predicts churn probability using machine learning models.

Business Problem Statement
Companies struggle to understand customer behavior patterns and identify early signs of churn. TEYZIX CORE requires a combined analytics and ML system that transforms raw customer data into actionable insights and predictions.

Task Requirements

1. **Data Analysis Module**
   Load dataset (CSV format)
   Perform exploratory data analysis
   Handle missing values and outliers
   Generate summary statistics
2. **Feature Engineering**
   Create behavioral features such as
   Usage frequency
   Tenure
   Payment history patterns
   Support interaction count
   Trend-based features

3. **Data Visualization**
   Revenue trends
   Customer segmentation charts
   Churn distribution
   Correlation heatmap
   Category-wise comparisons
4. **Machine Learning Model**
   Train at least 2 models
   Logistic Regression
   Random Forest / XGBoost

Evaluate using
Accuracy
Precision
Recall
F1 Score
ROC-AUC

5. **Customer Segmentation**
   Segment customers into
   High value
   Medium value
   Low value

Based on behavior + spending patterns

6. **Basic Prediction System**
   Predict churn probability
   Assign risk category
   Low / Medium / High risk
7. **Business Insights Report**
   Top churn reasons
   High-risk customer traits
   Revenue impact estimation

Technical Requirements
Python required
Use pandas, numpy, matplotlib/seaborn
Use scikit-learn for ML models
Modular notebook or scripts required

Expected Deliverables
Python notebook (.ipynb)
Dataset used
Visualizations

Model results report
Summary insights document

Evaluation Criteria
Data analysis 25%
Feature engineering 20%
Model performance 20%
Visualization quality 15%
Insights clarity 10%
Code structure 10%

Bonus Challenge
Streamlit dashboard
SHAP explainability
Automated weekly prediction pipeline
Email report generation